

# Columbia Econ Math Camp 2025

Static Optimization

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# 1. Optimization Theorems

## Definition 1 (Optimization problems)

Let  $(Y, \leq)$  be a poset,  $f : X \rightarrow Y$ ,  $D \subset X$ . A **maximization problem** takes the form

$$\max_{x \in X} f(x) \text{ s.t. } x \in D \quad \text{or} \quad \max_{x \in D} f(x)$$

where  $f$  is called the **objective function**,  $x$  the **choice variable**, and  $D$  the **constraint (or feasible) set**. We say  $x \in X$  is (a) **feasible (point)** iff  $x \in D$ . The problem is said to **have a solution** iff the set of **maximizers (or maximum points)** defined as

$$\arg \max_{x \in X} \{f(x) : x \in D\} := \{x^* \in D : f(x^*) \geq f(x) \forall x \in D\}$$

is non-empty. The **maximum (value)** of the problem is usually denoted  $f(x^*) := \max\{f(x) : x \in D\}$ .

## Proposition 1

Let  $f$  be a function from  $X$  to the poset  $(Y, \leq)$ , and let  $D \subset X$ . Let  $\{D_\alpha\}_{\alpha \in A}$  be a family of subsets of  $D$  such that  $\bigcup_{\alpha \in A} D_\alpha = D$ . For each  $\alpha \in A$ , let  $X_\alpha^* := \arg \max_{x \in X} \{f(x) : x \in D_\alpha\}$ . Suppose that  $X_\alpha^* \neq \emptyset$  for any  $\alpha \in A$ . Then

$$\arg \max_{x \in X} \{f(x) : x \in D\} = \arg \max_{x \in X} \left\{ f(x) : x \in \bigcup_{\alpha \in A} X_\alpha^* \right\}$$

Intuitively, this proposition says that when we optimize  $f$  over the set  $D$ , we can optimize within several subsets of  $D$ . As long as the union of subsets equals  $D$ , we can collect the maximizers over each piece and compare them.

# Optimization of real-valued functions

Henceforth, assume Euclidean metric spaces where not specified.

## Proposition 2

Let  $(X, d)$  be a metric space,  $f : X \rightarrow \mathbb{R}$ ,  $D \subset X$  is nonempty, and consider optimization problem

$$\max_{x \in X} f(x) \text{ s.t. } x \in D.$$

If  $D$  is compact and  $f$  is continuous, then the problem has a solution.

- This result is basically a re-statement of the Weierstrass theorem.

# Optimization of real-valued functions

Now some results about the shape of the maximizer set:

## Proposition 3

Let  $X$  be a set in real vector space  $(V, +, \cdot)$ ,  $f : X \rightarrow \mathbb{R}$  is quasiconcave,  $D \subset X$  is a convex set in  $V$ . Then  $\arg \max\{f(x) : x \in D\}$  is a convex set in  $V$ .

If we strengthen to strict q-conc, then we have uniqueness:

## Proposition 4

Let  $X$  be a set in real vector space  $(V, +, \cdot)$ ,  $f : X \rightarrow \mathbb{R}$  is strictly quasiconcave,  $D \subset X$  is a convex set in  $V$ . Then  $\arg \max\{f(x) : x \in D\}$  has at most one element.

# Optimization in $\mathbb{R}^n$

Henceforth  $X \subset \mathbb{R}^n$ , and let's focus on identifying necessary and sufficient conditions:

## Theorem 1 (First and second order conditions)

Let  $X \subset \mathbb{R}^n$ ,  $D \subset X$ ,  $f : X \rightarrow \mathbb{R}$ , and consider problem

$$\max_{x \in X} f(x) \text{ s.t. } x \in D.$$

with a maximizer  $\mathbf{x}^* \in \text{int}(D)$ . Then

- ① **(First order condition)** If  $f$  is differentiable at  $\mathbf{x}^*$ , then  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ .
- ② **(Second order condition)** If  $f$  is differentiable in an open ball around  $\mathbf{x}^*$ , and is twice differentiable at  $\mathbf{x}^*$ , then  $H_f(\mathbf{x}^*)$  is negative semi-definite.

# Optimization in $\mathbb{R}^n$

- The theorem describes your usual practice in economics: Any *interior* maximum point must satisfy FOC and SOC. Call this **type 1 candidates** for maximums
- There are two more cases though, call them **type 2** candidates:
  - ① Maximizers in the corners of  $D$  (i.e. outside  $\text{int}(D)$ ).
  - ② Maximizers at points where partials do not exist.
- In practice, we compare the value of  $f$  achieved by all type 1 and type 2 candidates, and pick whichever achieves the largest value of  $f$ .
- Note that if  $H_f(\mathbf{x}^*)$  is negative *definite*, then we talk about a **locally sufficient SOC**. This means that, together with the FOC, point  $\mathbf{x}^*$  is guaranteed to be a **strict local maximizer**.
- Formally, it is local in the sense that for points  $\mathbf{x}$  in some open ball  $B_\delta(\mathbf{x}^*)$  excluding the center, we have  $f(\mathbf{x}^*) > f(\mathbf{x})$ .
- Let's now look at conditions for *global* maximizers.



## Proposition 5

Let  $X$  be convex set in  $\mathbb{R}^n$ ,  $D \subset X$ ,  $f : X \rightarrow \mathbb{R}$  is concave. If  $f$  is differentiable  $\mathbf{x}^* \in \text{int}(X) \cap D$  and  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ , then

$$\mathbf{x}^* \in \arg \max \{f(\mathbf{x}) : \mathbf{x} \in D\}$$

If we only have q-conc, this does not hold. For example  $f(x) = x^3$  is q-conc but  $f$  does not have a maximum. In those cases instead we have:

## Proposition 6

Let  $X$  be convex set in  $\mathbb{R}^n$ ,  $D \subset X$ ,  $f : X \rightarrow \mathbb{R}$  is quasi-concave. If

- ①  $f$  is differentiable  $\mathbf{x}^* \in \text{int}(X) \cap D$  with  $\nabla f(\mathbf{x}^*) = \mathbf{0}$ , and
- ②  $f$  is  $C^2$  in some open ball around  $\mathbf{x}^*$  and  $H_f(\mathbf{x}^*)$  is negative definite.

Then  $\mathbf{x}^* \in \arg \max \{f(\mathbf{x}) : \mathbf{x} \in D\}$

## 2. Kuhn Tucker Optimization

# Optimization with inequalities

- In the remainder of this chapter, we will be interested **optimization problem in  $\mathbb{R}^n$  with inequalities**. These problems have the form

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \text{ s.t. } g(\mathbf{x}) \geq \mathbf{0}, h(\mathbf{x}) = \mathbf{0}$$

- Here  $X \subset \mathbb{R}^n$ ,  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$  ( $k$  inequalities) and  $h : X \rightarrow \mathbb{R}^m$  ( $m$  equalities).
- The **Lagrangian function** of this problem is

$$\begin{aligned} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &:= f(\mathbf{x}) + \boldsymbol{\lambda}^T g(\mathbf{x}) + \boldsymbol{\mu}^T h(\mathbf{x}) \\ &= f(\mathbf{x}) + \sum_{j=1}^k \lambda_j g_j(\mathbf{x}) + \sum_{l=1}^m \mu_l h_l(\mathbf{x}) \end{aligned}$$

where  $\lambda_j$ 's and  $\mu_l$ 's are the **Lagrangian multipliers**.

# Optimization with inequalities

Let's define what a constraint qualification is:

## Definition 2 (Constraint qualification)

Let  $X$  be an open set in  $\mathbb{R}^n$ , and  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$ ,  $h : X \rightarrow \mathbb{R}^m$  are all  $C^1$  functions. In the problem

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \text{ s.t. } g(\mathbf{x}) \geq \mathbf{0}, h(\mathbf{x}) = \mathbf{0}$$

we say that the inequality constraint  $g_j(\mathbf{x}) \geq 0$  is **binding** at feasible point  $\hat{\mathbf{x}} \in X$  if  $g_j(\hat{\mathbf{x}}) = 0$ . Moreover, we say that the **constraint qualification (CQ)** holds at  $\hat{\mathbf{x}}$  iff the gradients of all constraints binding at  $\hat{\mathbf{x}}$  are L.I. Otherwise, we say the **CQ fails at  $\hat{\mathbf{x}}$** .

Note that the equality constraints  $h_l$  are always trivially binding! So always add their gradients  $\nabla h_l(\hat{\mathbf{x}})$  to the set of vectors to be analyzed for linear dependence.

# Optimization with inequalities

With that defined, we can talk about the necessary conditions for a maximum, also known as the **Kuhn Tucker (necessary) conditions**:

## Theorem 2 (Kuhn-Tucker)

Let  $X$  be an open set in  $\mathbb{R}^n$ , and  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$ ,  $h : X \rightarrow \mathbb{R}^m$  are all  $C^1$  functions. In the problem

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \text{ s.t. } g(\mathbf{x}) \geq \mathbf{0}, h(\mathbf{x}) = \mathbf{0}$$

If  $\mathbf{x}^*$  is a maximizer of this problem, and CQ holds at  $\mathbf{x}^*$ , then there exists unique pair  $(\lambda^*, \mu^*) \in \mathbb{R}_+^k \times \mathbb{R}^m$  s.t. following conditions hold

- ❶ **First order (FOC):**  $\nabla f(\mathbf{x}^*) + (\lambda^*)^T g'(\mathbf{x}^*) + (\mu^*)^T h'(\mathbf{x}^*) = \mathbf{0}$
- ❷ **Complementary Slackness (CSC):**  $h(\mathbf{x}^*) = \mathbf{0}$  and for each  $j \in \{1, 2, \dots, k\}$  we have  $\lambda_j^* \geq 0$ ,  $g_j(\mathbf{x}^*) \geq 0$  and  $\lambda_j^* g_j(\mathbf{x}^*) = 0$ .

# Optimization with inequalities

Now let's look at necessary and sufficient conditions for optimums:

## Proposition 7

Let  $X$  be an open, convex set in  $\mathbb{R}^n$ ,  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$ , and  $h : X \rightarrow \mathbb{R}^m$  be  $C^1$  functions. Consider the problem

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \geq \mathbf{0} \text{ and } h(\mathbf{x}) = \mathbf{0}$$

Suppose  $\mathbf{x}^*$  is feasible, and there exists  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \in \mathbb{R}_+^k \times \mathbb{R}^m$  such that the following three conditions hold

- (1) FOC:  $\nabla f(\mathbf{x}^*) + (\boldsymbol{\lambda}^*)^T g'(\mathbf{x}^*) + (\boldsymbol{\mu}^*)^T h'(\mathbf{x}^*) = \mathbf{0}$ .
- (2) CSC:  $\lambda_j^* \geq 0$ ,  $g_j(\mathbf{x}^*) \geq 0$  and  $\lambda_j^* g_j(\mathbf{x}^*) = 0$  for  $j \in \{1, \dots, k\}$ .
- (3) The Lagrangian on  $\mathbf{x}$ , i.e.  $\mathcal{L}_{\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*}(\mathbf{x}) := \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ , is concave,

then  $\mathbf{x}^*$  is a maximizer of this problem.

# Optimization with inequalities

Let's look at a version with concavity on  $f$

## Proposition 8

Let  $X$  be open, convex set in  $\mathbb{R}^n$ ,  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$ ,  $h : X \rightarrow \mathbb{R}^m$  are  $C^1$  functions,  $f$  is concave. Consider the problem

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \geq \mathbf{0} \text{ and } h(\mathbf{x}) = \mathbf{0}$$

If  $\mathbf{x}^*$  is feasible, and there exists  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \in \mathbb{R}_+^k \times \mathbb{R}^m$  such that

- (1) FOC:  $\nabla f(\mathbf{x}^*) + (\boldsymbol{\lambda}^*)^T g'(\mathbf{x}^*) + (\boldsymbol{\mu}^*)^T h'(\mathbf{x}^*) = \mathbf{0}$
- (2) CSC:  $\lambda_j^* \geq 0$ ,  $g_j(\mathbf{x}^*) \geq 0$ , and  $\lambda_j^* g_j(\mathbf{x}^*) = 0$  for  $j \in \{1, \dots, k\}$ ,
- (3)  $\lambda_j^* g_j$  is quasi-concave for  $j = 1, \dots, k$ , and  $\mu_l^* h_l$  is quasi-concave for  $l = 1, \dots, m$ ,

then  $\mathbf{x}^*$  is a maximizer of this problem.

# Optimization with inequalities

Finally, let's look at a version with only q-conc on  $f$

## Proposition 9

Let  $X$  be open, convex set in  $\mathbb{R}^n$ ,  $f : X \rightarrow \mathbb{R}$ ,  $g : X \rightarrow \mathbb{R}^k$ ,  $h : X \rightarrow \mathbb{R}^m$  are  $C^1$  functions,  $f$  is quasi-concave. Consider

$$\max_{\mathbf{x} \in X} f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \geq \mathbf{0} \text{ and } h(\mathbf{x}) = \mathbf{0}$$

If  $\mathbf{x}^*$  is feasible, and there exists  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \in \mathbb{R}_+^k \times \mathbb{R}^m$  such that

- (1) FOC:  $\nabla f(\mathbf{x}^*) + (\boldsymbol{\lambda}^*)^T g'(\mathbf{x}^*) + (\boldsymbol{\mu}^*)^T h'(\mathbf{x}^*) = \mathbf{0}$
- (2) CSC:  $\lambda_j^* \geq 0$ ,  $g_j(\mathbf{x}^*) \geq 0$ , and  $\lambda_j^* g_j(\mathbf{x}^*) = 0$  for  $j \in \{1, \dots, k\}$ ,
- (3)  $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$ ,  $\lambda_j^* g_j$  is quasi-concave for  $j = 1, \dots, k$ , and  $\mu_l^* h_l$  is quasi-concave for  $l = 1, \dots, m$ ,

then  $\mathbf{x}^*$  is a maximizer of this problem.



# Optimization with inequalities

Below is a summary with the basic algorithm you need to follow to solve a mixed optimization problem with KKT:

- 1 Set up the Lagrangean and find every point  $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$  that satisfy the necessary KT conditions (i.e. FOC and CSC). Denote the set of first elements from these triples as  $X^*$
- 2 Find maximizers across points in which the CQ fails. Denote this second set of maximizers as  $X^{**}$
- 3 Discard any of the points  $\mathbf{x}^*$  and  $\mathbf{x}^{**}$  that are not feasible (i.e. do not satisfy the restrictions), from  $X^*$  and  $X^{**}$ .
- 4 Find  $\mathbf{x}^{***}$  s.t.  $f(\mathbf{x}^{***}) = \max\{f(\mathbf{x}) : \mathbf{x} \in X^* \cup X^{**}\}$ .
- 5 Justify that  $\mathbf{x}^{***}$  is the maximizer using some theorem or, in the absence of a theorem at hand, one of the sufficient conditions on convexity of the functions.

# Optimization with inequalities

Now let's look at the famous Envelope Theorem

## Theorem 3 (Envelope)

Let  $X$  be open set in  $\mathbb{R}^n$ ,  $A$  is an open set of **parameters** in  $\mathbb{R}^s$ ,  $f : X \times A \rightarrow \mathbb{R}$ ,  $g : X \times A \rightarrow \mathbb{R}^k$ , and  $h : X \times A \rightarrow \mathbb{R}^m$  are  $C^1$  functions. For each  $\alpha \in A$ , define **parametrized problem**  $P(\alpha)$  as

$$\max_{\mathbf{x} \in X} f(\mathbf{x}, \alpha) \quad \text{s.t.} \quad g(\mathbf{x}, \alpha) \geq 0 \text{ and } h(\mathbf{x}, \alpha) = 0$$

Let  $\hat{A} := \{\alpha \in A : P(\alpha) \text{ has solution}\}$ , and define  $f^* : \hat{A} \rightarrow \mathbb{R}$  as

$$f^*(\alpha) := \max_{\mathbf{x} \in X} \{f(\mathbf{x}, \alpha) : g(\mathbf{x}, \alpha) \geq 0 \text{ and } h(\mathbf{x}, \alpha) = 0\}$$

For parameter  $\alpha^* \in A$ , suppose:

- (1)  $P(\alpha^*)$  has a unique maximizer  $\mathbf{x}^*$ , and CQ holds at  $\mathbf{x}^*$ .
- (2)  $\exists \varepsilon, r > 0$  s.t.  $\forall \alpha \in B_\varepsilon(\alpha^*)$ ,  $(\arg \max P(\alpha)) \cap B_r(\mathbf{x}^*) \neq \emptyset$ .

# Optimization with inequalities

Now let's look at the famous Envelope Theorem

## Theorem 3 (Envelope - cont'd)

If all these conditions hold, then the value function  $f^*$  is differentiable at  $\alpha^*$ , and

$$\begin{aligned} \left. \frac{d}{d\alpha} f^*(\alpha) \right|_{\alpha=\alpha^*} &= \left. \frac{d}{d\alpha} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*, \alpha) \right|_{\alpha=\alpha^*} \\ &= \left. \frac{d}{d\alpha} f(\mathbf{x}^*, \alpha) \right|_{\alpha=\alpha^*} + \boldsymbol{\lambda}^{*T} \left. \frac{d}{d\alpha} g(\mathbf{x}^*, \alpha) \right|_{\alpha=\alpha^*} + \boldsymbol{\mu}^{*T} \left. \frac{d}{d\alpha} h(\mathbf{x}^*, \alpha) \right|_{\alpha=\alpha^*} \end{aligned}$$

where  $\mathbf{x}^*$  and  $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$  are the unique Lagrangian multipliers found by Kuhn Tucker theorem for the problem  $P(\alpha^*)$ .